```
Name - Sanskruti Gorle
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Batch - A1



%{
#include "y.tab.h"
%}

%%
"for"  { return FOR; }
[a-zA-Z_][a-zA-Z0-9_]+  { return ID; }
[0-9]+ { return NUM; }
"<="|">="|"=="|"!="|"<"|">" { return RELOP; }
"++"  { return INC; }
"--"   { return DEC; }
[ \t\n]+  ;
.   { return yytext[0]; }
%%

int yywrap()
{
    return 1;
}
```

```
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%{
#include <stdio.h>

int yylex(void);
int yyerror(const char *s);
%}

%token FOR ID NUM RELOP INC DEC

%%
stmt : FOR '(' ID '=' NUM ';' ID RELOP NUM ';' ID INC ')'
     | FOR '(' ID '=' NUM ';' ID RELOP NUM ';' ID DEC ')'
     | ID RELOP ID '?' ID ':' ID
     | ID RELOP NUM '?' ID ':' ID
     | ID RELOP ID '?' NUM ':' NUM
     | ID RELOP NUM '?' NUM ':' NUM
     ;
%%

int yyerror(const char *s)
{
    return 1;
}

int main()
{
    printf("Enter a for loop or ternary expression:\n");

    if(yyparse() == 0)
        printf("Valid statement\n");
    else
        printf("Invalid statement\n");

    return 0;
}
```

```
sanskruti@LAPTOP-ICU7078D:~$ gedit cc10.y
sanskruti@LAPTOP-ICU7078D:~$ yacc -d cc10.y
sanskruti@LAPTOP-ICU7078D:~$ flex cc10.l
sanskruti@LAPTOP-ICU7078D:~$ gcc y.tab.c lex.yy.c -ll
sanskruti@LAPTOP-ICU7078D:~$ ./a.out
Enter a for loop or ternary expression:
for(i=0;i<5;i++)
Valid statement
sanskruti@LAPTOP-ICU7078D:~$ ./a.out
Enter a for loop or ternary expression:
a>b?a:b
Valid statement
sanskruti@LAPTOP-ICU7078D:~$ ./a.out
Enter a for loop or ternary expression:
a>10?20:30
Valid statement
sanskruti@LAPTOP-ICU7078D:~$ ./a.out
Enter a for loop or ternary expression:
for(i=0;i<5;i+)
Invalid statement
sanskruti@LAPTOP-ICU7078D:~$ ./a.out
Enter a for loop or ternary expression:
a>b?a:b
Valid statement
sanskruti@LAPTOP-ICU7078D:~$ ./a.out
Enter a for loop or ternary expression:
a>b?a
Invalid statement
```

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